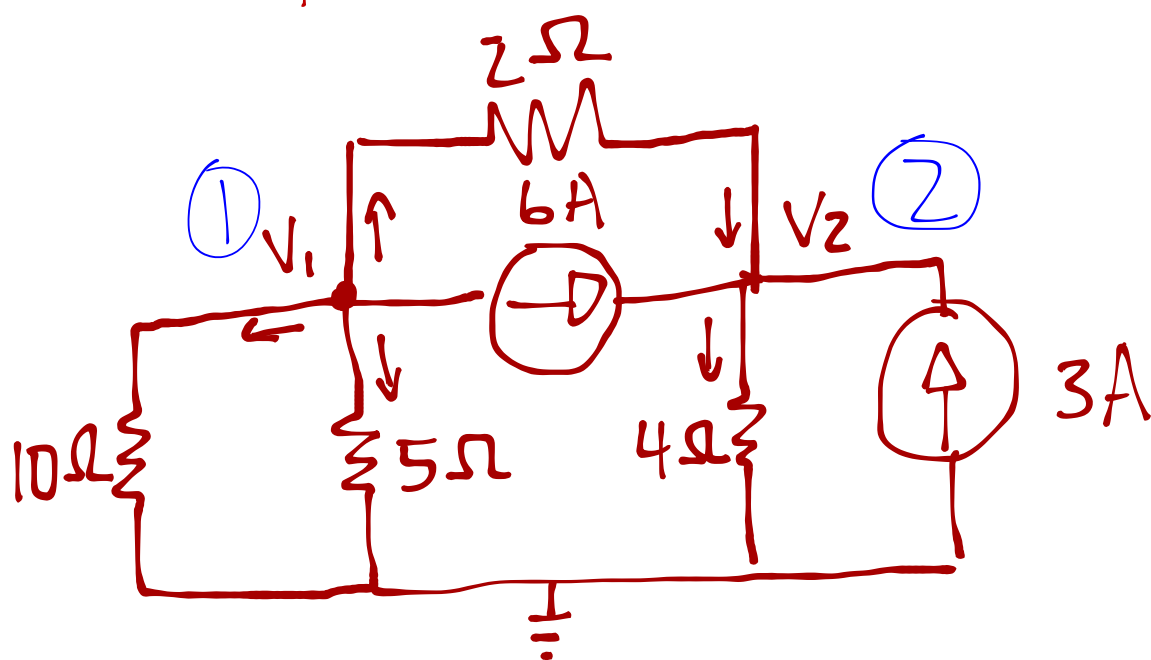


Tutorial 3 Memo

24 Feb 2017

3.2 Find V_1 and V_2



@ Node ①: $\sum I_{in} = \sum I_{out}$

$$0 = \frac{V_1 - 0}{10} + \frac{V_1 - 0}{5} + \frac{V_1 - V_2}{2} + 6 \rightarrow \text{Multiply with 10}$$

$$\therefore V_1 + 2V_1 + 5V_1 - 5V_2 = -60 \quad \text{↳ Multiply - out}$$

$$-8V_1 + 5V_2 = 60$$

$$-8V_1 = 60 - 5V_2 \dots \textcircled{1}$$

@ Node ②: $\frac{V_1 - V_2}{2} + 3 + 6 = \frac{V_2}{4} \rightarrow \text{Multiply with 4}$

$$2V_1 - 2V_2 + (9 \times 4) = V_2 \rightarrow \text{Multiply with -4}$$

$$4(-2V_1 + 3V_2 = 36)$$

$$-8V_1 = 144 - 12V_2 \dots \textcircled{2}$$

Solve using Simultaneous equations

$$\textcircled{1} = \textcircled{2}$$

$$60 - 5V_2 = 144 - 12V_2$$

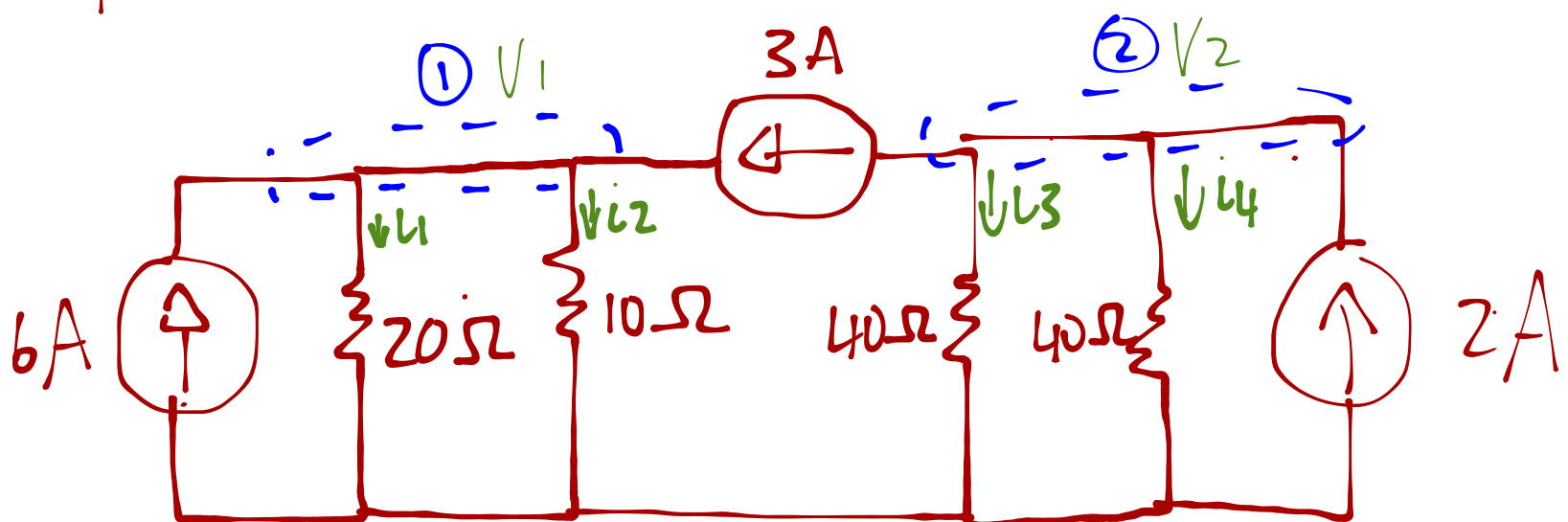
$$\therefore V_2 = 12V$$

Substitute V_2 into $\textcircled{1}$

$$-8V_1 = 60 - 5(12)$$

$$\therefore V_1 = 0V$$

3.4 solve for i_1 to i_4



@Node ①: $\sum I_{in} = \sum I_{out}$

$$6 + 3 = \frac{V_1}{20} + \frac{V_1}{10} \rightarrow \text{Multiply with } 20$$

$$180 = V_1 + 2V_1$$

$$\therefore V_1 = 60V$$

$$i_1 = \frac{V_1}{20} = \frac{60}{20} = 3A \rightarrow$$

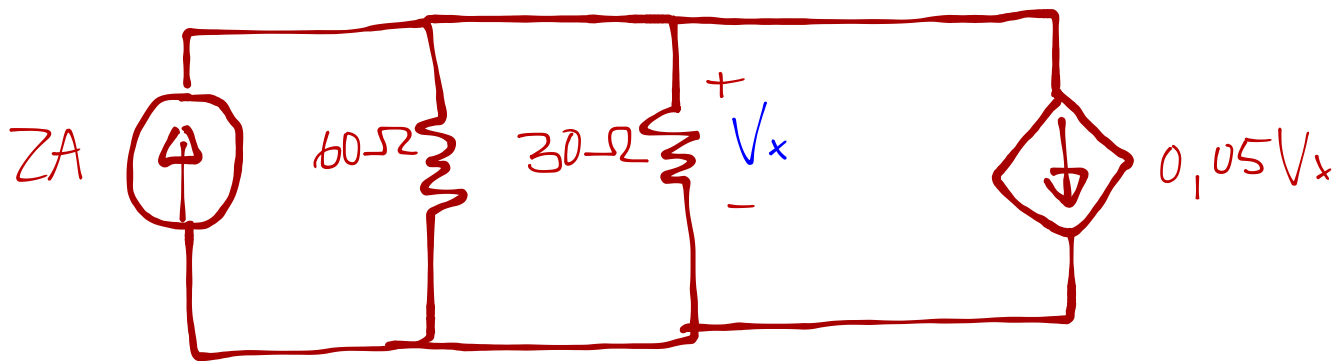
$$i_2 = \frac{V_1}{10} = \frac{60}{10} = 6A \rightarrow$$

@Node ②: $2 = 3 + i_3 + i_4$

$$2 - 3 = \frac{V_2}{40} + \frac{V_2}{40} \rightarrow \text{Multiply with } 20$$

$$-20 = V_2$$

$$\therefore i_3 = i_4 = \frac{V_2}{40} = \frac{-20}{40} = -0,5A \rightarrow$$

3.7 Solve for V_x 

1 node with voltage = V_x

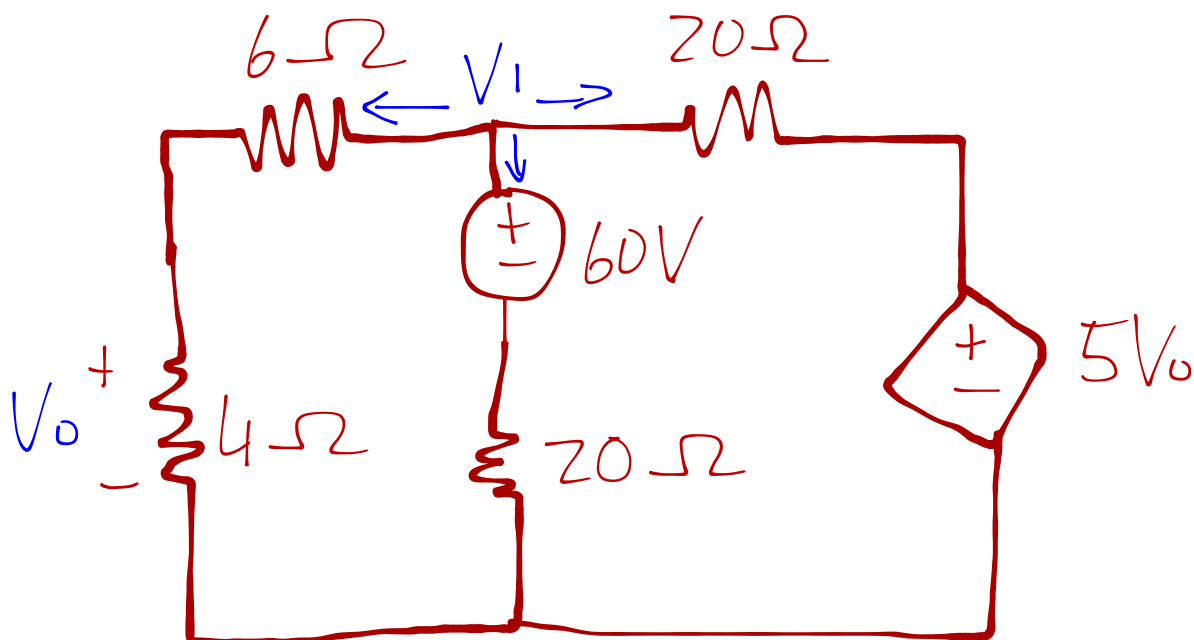
Nodal analysis: $\sum I_{in} = \sum I_{out}$

$$2 = \frac{V_x}{30} + \frac{V_x}{60} + 0.05V_x \rightarrow \text{Multiply with } 60$$

$$120 = V_x + 2V_x + 3V_x$$

$$120 = 6V_x$$

$$\therefore \underline{V_x = 20V}$$

3.8 Find V_o 

Nodal analysis @ V_1

$$\sum I_{in} = \sum I_{out}$$

$$0 = \underbrace{\frac{V_1 - 0}{6+4}}_{\text{series resistances}} + \frac{(V_1 - 60) - 0}{20} + \frac{V_1 - 5V_0}{20} \rightarrow \text{Multiply with } 20$$

$$R_s = R_1 + R_2$$

Used $I = \frac{V_{from} - V_{to}}{R}$

$$0 = 2V_1 + V_1 - 60 + V_1 - 5V_0$$

$$60 = 4V_1 - 5V_0 \dots \textcircled{1}$$

Voltage division:

$$V_0 = \left(\frac{4}{R_s} \right) V_1$$

$$V_0 = \left(\frac{4}{4+6} \right) V_1$$

$$= \frac{2}{5} V_1 \dots \textcircled{2}$$

∴ Substitute $\textcircled{2}$ in $\textcircled{1}$

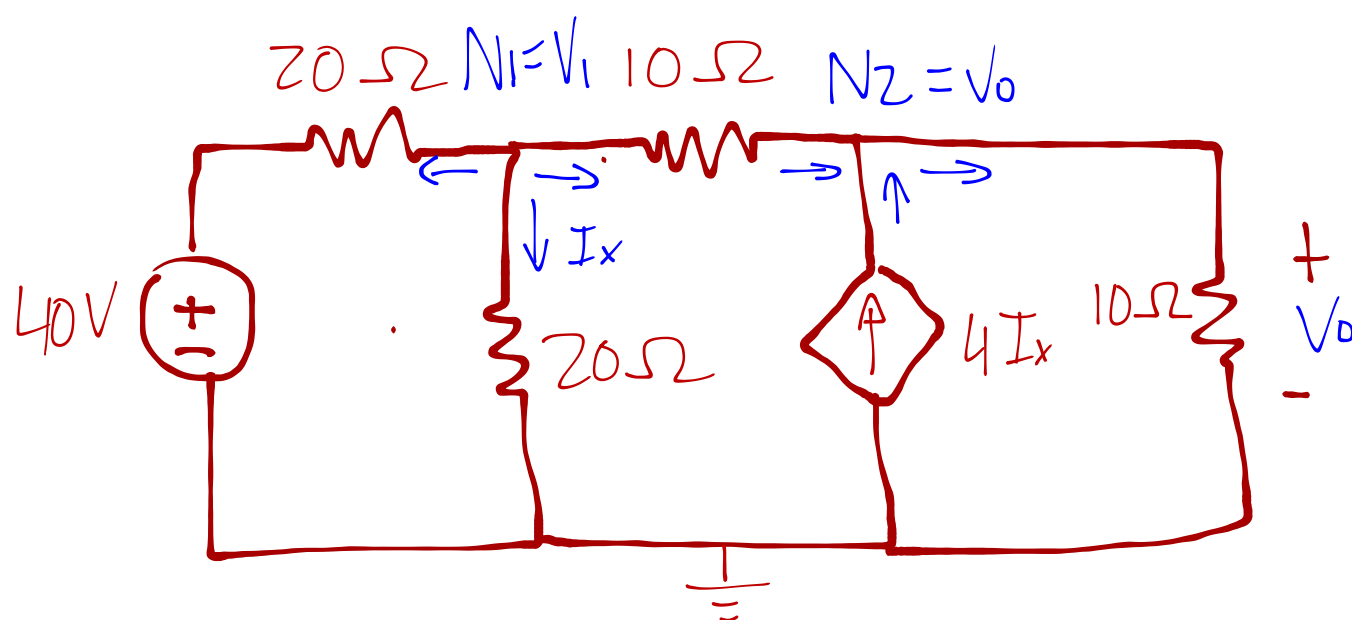
$$\Rightarrow 4V_1 - 5\left(\frac{2}{5}V_1\right) = 60$$

$$V_1 = 30V$$

Substitute V_1 in $\textcircled{2}$

$$\therefore V_0 = \frac{2}{5}(30) = \underline{12V} \rightarrow$$

3.12 Determine V_o



① Node 1: $\sum I_{in} = \sum I_{out}$

$$0 = \frac{V_1 - 40}{20} + \frac{V_1 - 0}{20} + \frac{V_1 - V_0}{10} \rightarrow \text{Multiply with 20}$$

$$0 = V_1 - 40 + V_1 + 2V_1 - 2V_0$$

$$40 = 4V_1 - 2V_0 \dots \textcircled{1}$$

② Node 2: $4I_x + \frac{V_1 - V_0}{10} = \frac{V_0 - 0}{10} \rightarrow \text{Multiply with 10}$

$$40I_x + V_1 = V_0 + V_0 \dots \textcircled{2}$$

Ohms law: $I = \frac{V}{R}$

$$\therefore I_x = \frac{V_1}{20} \dots \textcircled{3}$$

Substitute ③ into ②

$$2V_1 + V_1 = 2V_0$$

$$3V_1 = 2V_0$$

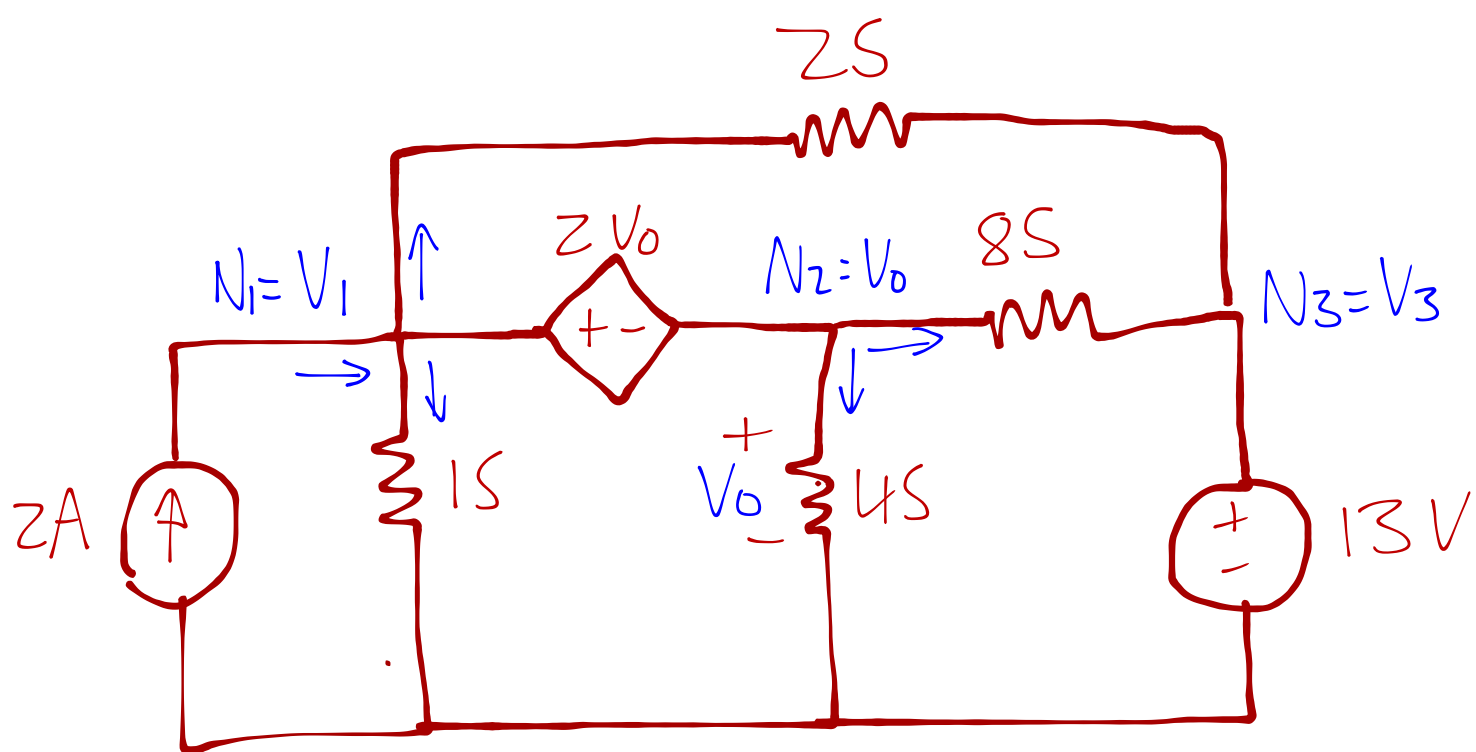
$$\therefore V_1 = \frac{2}{3} V_0 \dots \textcircled{4}$$

Substitute ④ into ①

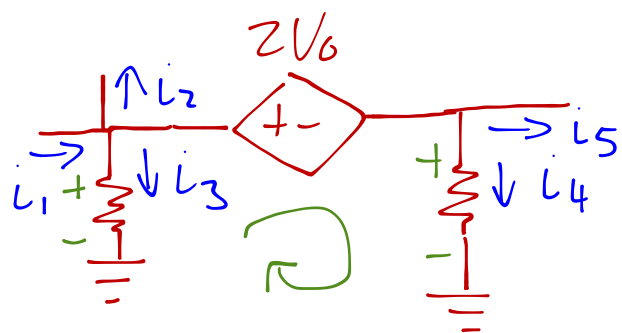
$$40 = 4\left(\frac{2}{3} V_0\right) - 2V_0$$

$$\therefore \underline{V_0 = 60} \quad \triangleright$$

3.16 Find V_1 to V_3



See the conjunction of N_1 and N_2 as a
Super node



Note !!!

$$\frac{1}{s} = R$$

KCL as a single node

$$\sum I_{in} = \sum I_{out}$$

$$2 = 1(V_1 - 0) + 2(V_1 - V_3) + 4(V_0 - 0) + 8(V_0 - V_3)$$

$$2 = 3V_1 + 12V_0 - 10V_3 \dots \textcircled{1}$$

KVL @ super node

$$\sum V = 0$$

$$-V_1 + 2V_0 + V_0 = 0$$

$$\therefore V_1 = 3V_0 \dots \textcircled{2}$$

Note $\underline{V_3 = 13V} \dots \textcircled{3}$

Substitute ③ and ② into ①

$$2 = 3V_1 + 12\left(\frac{1}{3}V_1\right) - 10(13)$$

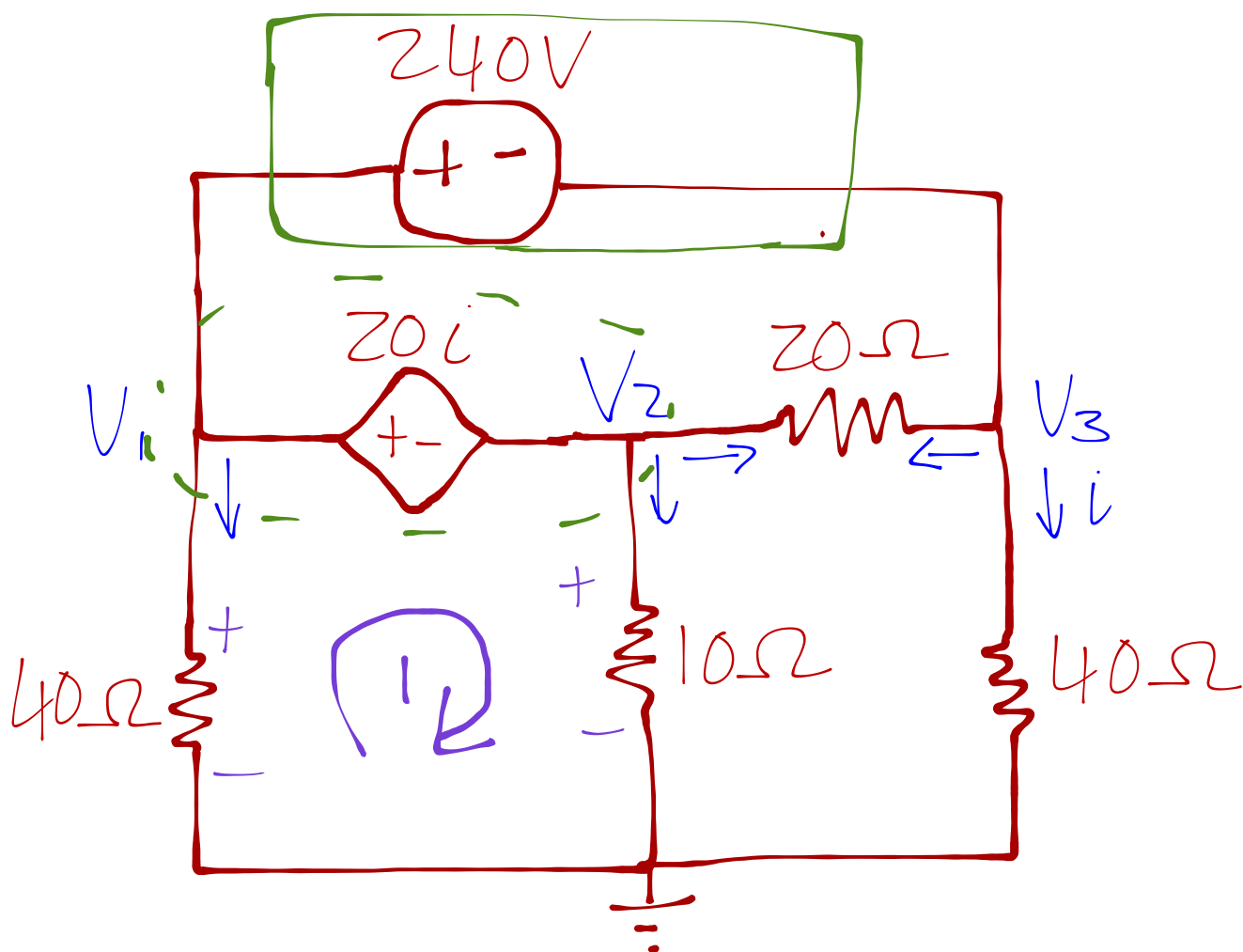
$$\therefore \underline{V_1 = 18.857V}$$

Substitute V_1 into ②

$$\frac{1}{3}(18.857) = V_0$$

$$\therefore V_0 = V_2 = \underline{6.286V}$$

3.20 Find V_1 to V_3



See the conjunction of Node 1 and Node 2
as a supernode

Note between Node 1 and Node 3 is
another supernode. For KCL we see
Node 1, 2 and 3 as one node.

$$\sum I_{in} = \sum I_{out}$$

$$0 = \frac{V_1 - 0}{40} + \frac{V_2 - 0}{10} + \frac{V_3 - 0}{40} + \cancel{\frac{V_2 - V_3}{20}} + \cancel{\frac{V_3 - V_2}{20}}$$

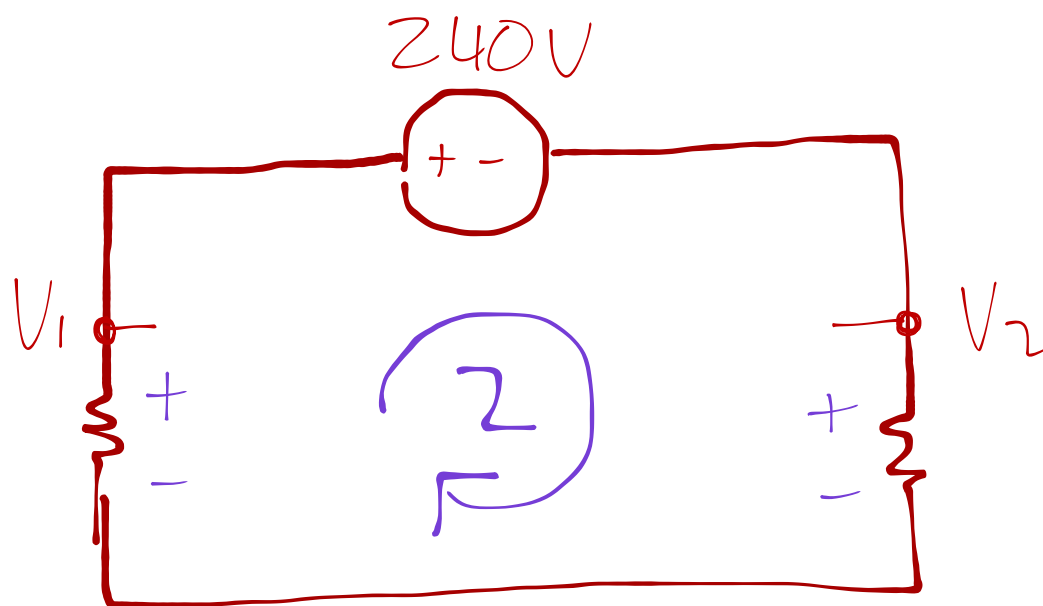
→ Multiply with 40

$$0 = V_1 + 4V_2 + V_3 \dots \textcircled{1}$$

KVL: $\sum V = 0$

$$-V_1 + 20i + V_2 = 0$$

$$\therefore V_1 - V_2 = 20i \dots \textcircled{2}$$



KVL2: $-V_1 + 240 + V_3 = 0$

$\therefore V_1 - V_3 = 240 \dots \textcircled{3}$

Ohms law $i = \frac{V}{R}$

$\therefore i = \frac{V_3}{40} \dots \textcircled{4}$

Substitute $\textcircled{4}$ into $\textcircled{3}$

$$V_1 - 40i = 240$$

$$V_1 - 240 = 40i$$

$$\frac{1}{2}(V_1 - 240) = 20i \dots \textcircled{5}$$

$$\text{Set } \textcircled{5} = \textcircled{2}$$

$$V_1 - V_2 = \frac{1}{2} (V_1 - 240)$$

$$V_2 = V_1 - \frac{1}{2} V_1 + 120 \dots \textcircled{6}$$

Substitute $\textcircled{6}$ and $\textcircled{3}$ into $\textcircled{1}$

$$V_1 + 4\left(\frac{1}{2} V_1 + 120\right) + (V_1 - 240) = 0$$

$$4 V_1 = -240$$

$$\underline{V_1 = -60 \text{ V} \rightarrow}$$

Substitute V_1 into $\textcircled{6}$ and $\textcircled{3}$

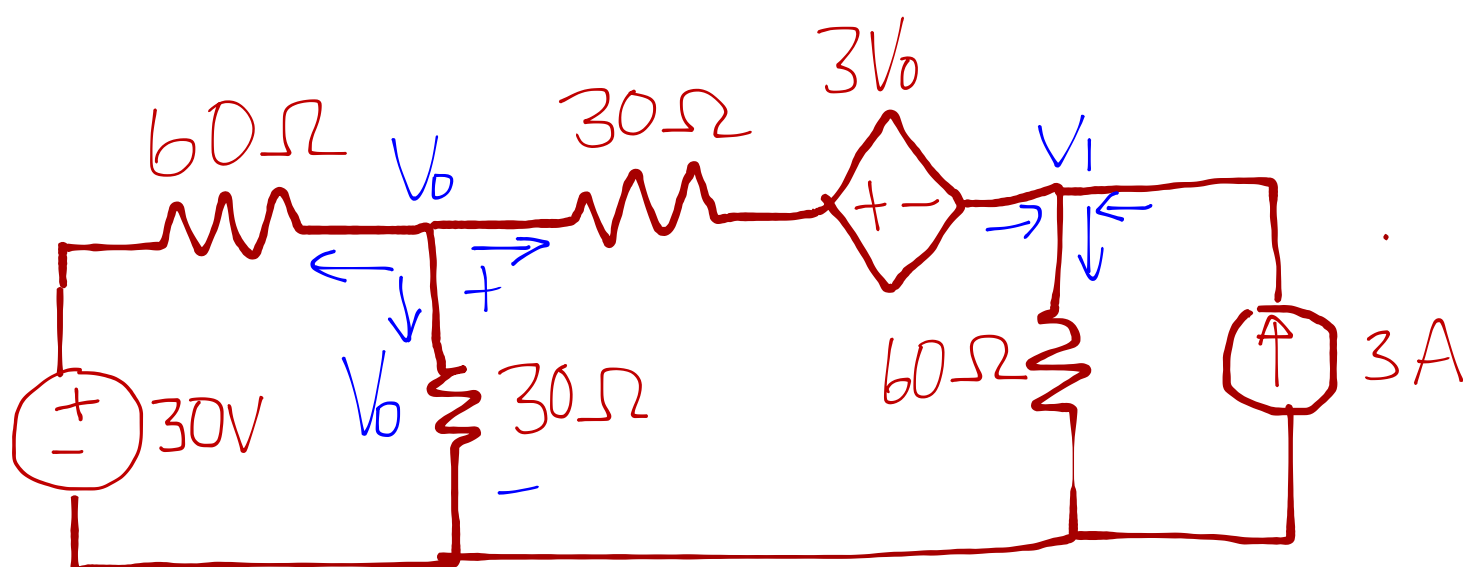
$$\textcircled{3}: -60 - 240 = V_3$$

$$\therefore \underline{V_3 = -300 \text{ V} \rightarrow}$$

$$\textcircled{6}: \frac{1}{2} (-60) + 120 = V_2$$

$$\therefore \underline{V_2 = 90 \text{ V} \rightarrow}$$

3.23 Find V_o



@ Node 0: $\sum I_{in} = \sum I_{out}$

$$0 = \frac{V_o - 30}{60} + \frac{V_o - 0}{30} + \frac{V_o - (3V_o + V_1)}{30}$$

→ Multiply with 60

$$0 = V_o - 30 + 2V_o + 2V_o - 6V_o - 2V_1$$

$$30 = -V_o - 2V_1$$

$$\therefore V_o = -30 - 2V_1 \dots \textcircled{1}$$

@ Node 1: $\frac{V_o - (3V_o + V_1)}{30} + 3 = \frac{V_1}{60}$ → Multiply with 60

$$2V_o - 6V_o - 2V_1 + 180 = V_1$$

$$-4V_o - 3V_1 = -180 \dots \textcircled{2}$$

Substitute ① in ②

$$-4(-30 - 2V_1) - 3V_1 = -180$$

$$120 + 8V_1 - 3V_1 = -180$$

$$5V_1 = -300$$

$$V_1 = -60V$$

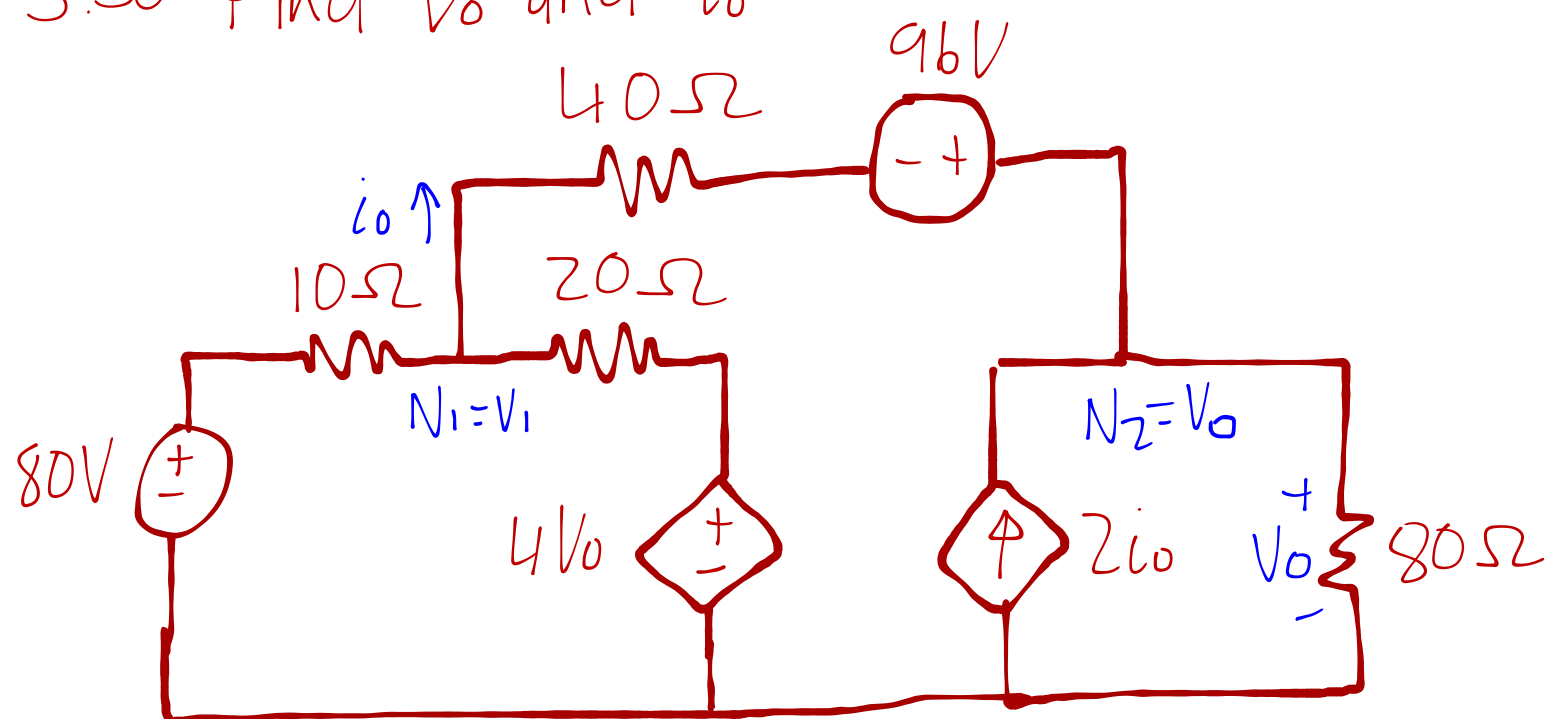
Substitute V_1 into ①

$$\therefore V_0 = -30 - 2(-60)$$

$$= 90V$$

→

3.30 Find V_0 and i_0



@ Node 1: $\sum I_{in} = \sum I_{out}$

$$\frac{80 - V_1}{10} + \frac{4V_0 - V_1}{20} = \frac{V_1 - (V_0 - 96)}{40} \quad \rightarrow \text{Multiply with 40}$$

$$320 - 4V_1 + 8V_0 - 2V_1 = V_1 - V_0 + 96$$

$$224 = 7V_1 - 9V_0 \dots \textcircled{1}$$

@ Node 2: $i_0 + 2i_0 = \frac{V_0 - 0}{80} \dots \textcircled{A}$

Replace i_0 with $\frac{V_1 - (V_0 - 96)}{40}$

$$\therefore 3\left(\frac{V_1 + 96 - V_0}{40}\right) = \frac{V_0}{80}$$

$$6V_1 + 576 - 6V_0 = V_0$$

$$6V_1 - 7V_0 = -576 \dots \textcircled{2}$$

Use Cramers rule to solve equation $\textcircled{2}$ and $\textcircled{3}$ simultaneously

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_0 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 7 & -9 \\ 6 & -7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_0 \end{bmatrix} = \begin{bmatrix} 224 \\ -576 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 7 & -9 \\ 6 & -7 \end{vmatrix} = (7)(-7) - (-9)(6) = -49 + 54$$

$$\Delta = 5$$

→ replace $\hat{y}$ into 1st column

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} 224 & -9 \\ -576 & -7 \end{vmatrix} = 224(-7) - (-9)(-576) \\ &= -1568 - 5184 \\ &= -6752 \end{aligned}$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 7 & 224 \\ 6 & -576 \end{vmatrix} = 7(-576) - (224)(6) \\ &= -4032 - 1344 \\ &= -5376 \end{aligned}$$

using determinants V_1 and V_0 can be calculated

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{-6752}{5} = \underline{-1350,4V}$$

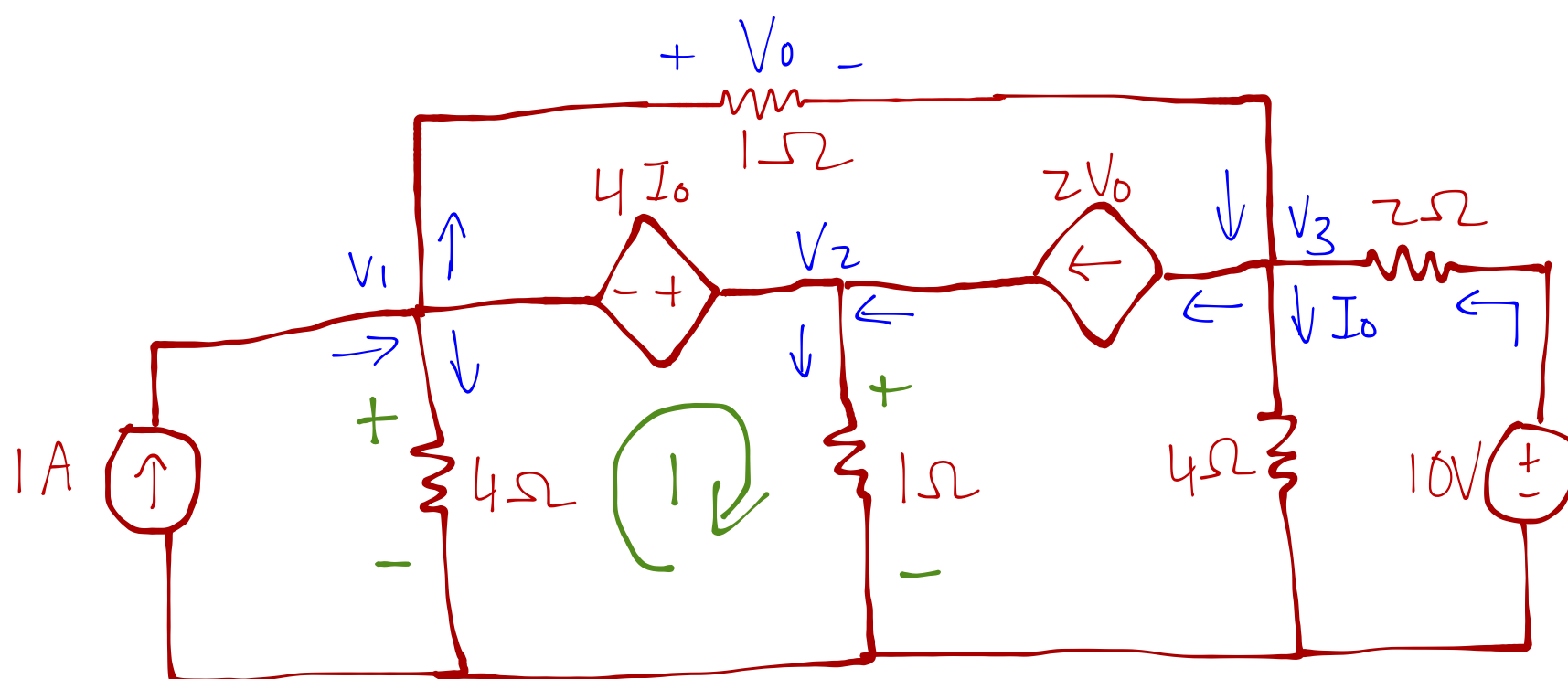
$$V_0 = \frac{\Delta_2}{\Delta} = \frac{-5376}{5} = \underline{-1075,2V}$$

substitute V_0 into (A)

$$3i_0 = \frac{-1075,2}{80}$$

$$\therefore i_0 = \underline{-4,48A}$$

3.31 Find V_1 to V_3



Start by finding equations from the Supernode between Node 1 and 2

KCL: $\sum I_{in} = \sum I_{out}$

$$1 + 2V_0 = \frac{V_1 - 0}{4} + \frac{V_1 - V_3}{1} + \frac{V_2 - 0}{1} \dots \textcircled{1}$$

Note V_0 is between V_1 and V_3

$$\therefore V_0 = V_1 - V_3 \dots \textcircled{2}$$

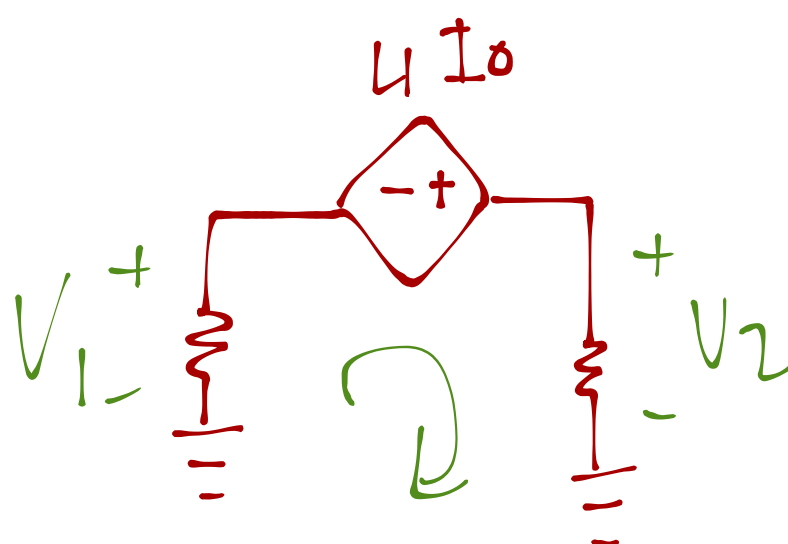
Substitute $\textcircled{2}$ into $\textcircled{1}$

$$1 + 2(V_1 - V_3) = \frac{V_1}{4} + V_1 - V_3 + V_2 \rightarrow \text{Multiply with } 4$$

$$4 + 8V_1 - 8V_3 = V_1 + 4V_1 - 4V_3 + 4V_2$$

$$4 = -3V_1 + 4V_2 + 4V_3 \dots \textcircled{3}$$

KVL1:



$$\Rightarrow \sum V = 0$$

$$-V_1 - 4I_0 + V_2 = 0$$

$$\therefore V_2 = V_1 + 4I_0$$

Note using Ohms law $I_0 = \frac{V_3}{4} \dots \textcircled{4}$

$$\therefore V_2 = V_1 + V_3 \dots \textcircled{5}$$

@ Node 3: $\sum I_{in} = \sum I_{out}$

$$\frac{V_1 - V_3}{1} + \frac{10 - V_3}{2} = 2V_0 + \frac{V_3 - 0}{4} \rightarrow \text{Multiply with 4}$$

$$4V_1 - 4V_3 + 20 - 2V_3 = 8V_0 + V_3$$

Substitute (2) into equation

$$4V_1 - 7V_3 + 20 = 8(V_1 - V_3)$$

$$V_3 = 4V_1 - 20 \dots (6)$$

Substitute (5) into (3)

$$4 = -3V_1 + 4(V_1 + V_3) + 4V_3 \dots (7)$$

Substitute (6) into (7)

$$4 = V_1 + 4(4V_1 - 20) + 4(4V_1 - 20)$$

$$164 = 33V_1$$

$$\therefore V_1 = \underline{4.97V} \rightarrow$$

Substitute V_1 into (6)

$$V_3 = 4(4,97) - 20$$

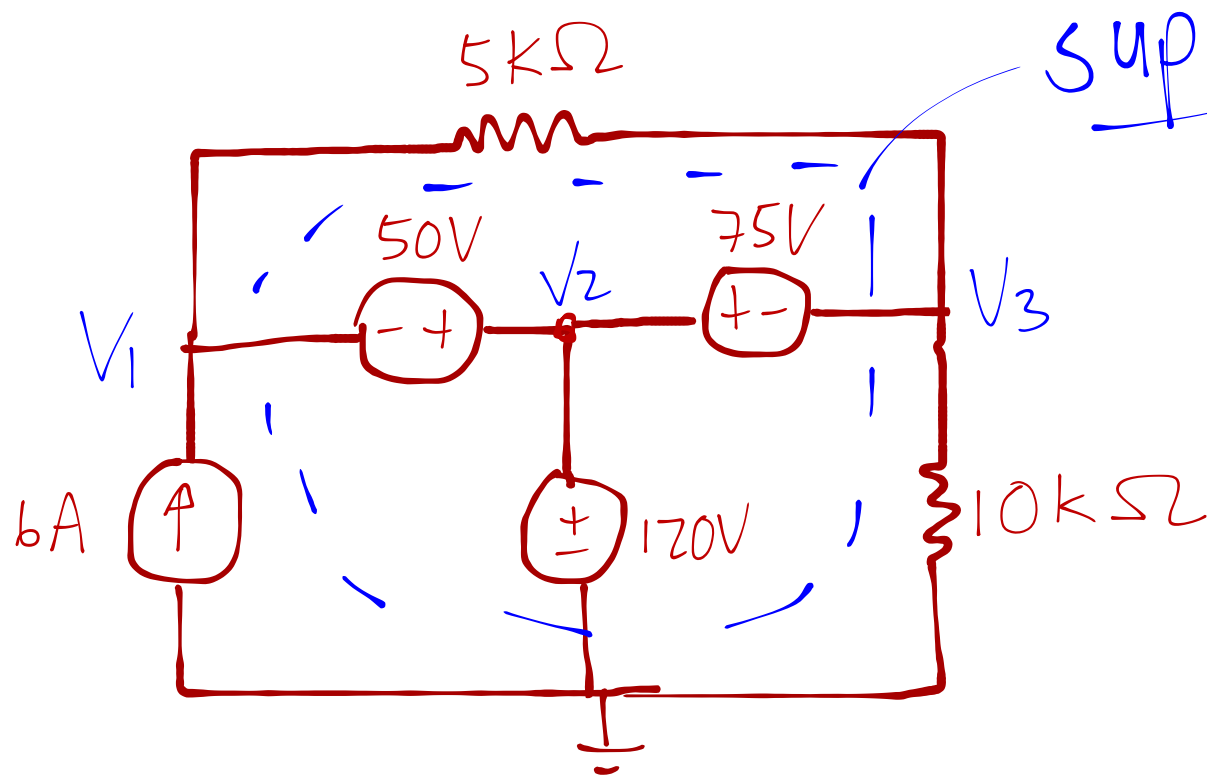
$$\therefore \underline{V_3 = -0,12V}$$

Substitute V_1 and V_3 into (5)

$$V_2 = 4,97 - 0,12$$

$$\therefore \underline{V_2 = 4,85V}$$

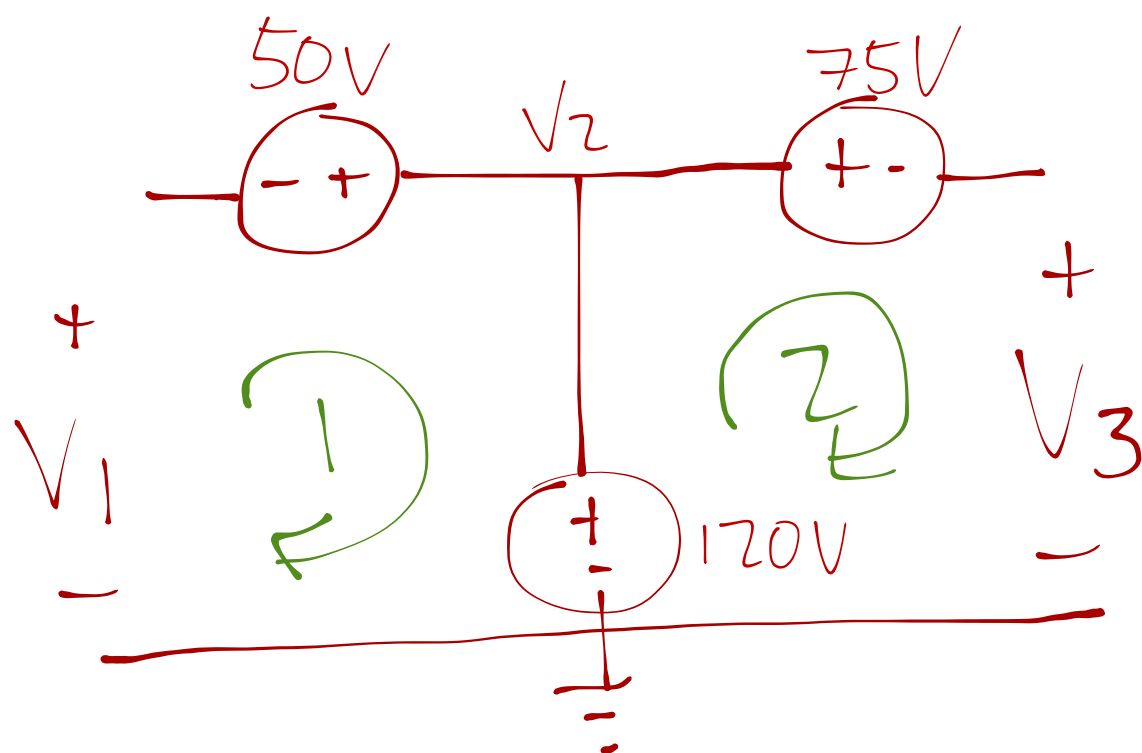
3.32 Find V_1 to V_3



The middle section of the circuit
forms one big supernode

The supernode has 2 independant loops
which both could be used with

KVL for equations



$$\text{KVL } \textcircled{1} : \sum V = 0$$

$$-V_1 - 50V + 120V = 0$$

$$\therefore \underline{V_1 = 70V} \rightarrow$$

Note $V_2 = 120V$ by visual inspection

$$\text{KVL (2)}: -120 + 75 + V_3 = 0$$

$$V_3 = 45\text{V}$$

—————>
